

UdjatTrack

API Design

v2.0: Final Implementation Reference

1. Base Configuration

Field	Value
Base URL	http://localhost:8080/api/v1
Authentication	Bearer JWT – include in every protected request
Content-Type	application/json
Token Expiry	Access Token: 15 min Refresh Token: 7 days
Pagination	?page=0&size=20&sort=createdAt,desc

1.1 Roles

Role	Description
SUPER_MANAGER	Platform owner. Manages fleet manager accounts and subscriptions.
FLEET_MANAGER	Manages drivers, vehicles, and trips for their fleet.
DRIVER	Operates trips via the mobile app.

1.2 Standard Response Envelope

All responses follow this envelope:

```
{
  "success": true | false,
  "data": { ... } | [ ... ] | null,
  "message": "Optional info or error description",
  "errors": [ "Validation error detail" ] // present on 400
}
```

1.3 Standard Error Codes

Code	Meaning
TRIP_NOT_FOUND	No trip matches the given ID.
DRIVER_NOT_ASSIGNED	The driver has no active vehicle assignment.
INVALID_STATE_TRANSITION	Trip state change is not allowed (e.g. END before START).
UNAUTHORIZED_ROLE	The caller's role does not permit this action.
TELEMETRY_INVALID	Telemetry payload is missing required fields or out of range.
VEHICLE_ALREADY_ON_TRIP	The vehicle is already assigned to an active trip.
DRIVER_ALREADY_ON_TRIP	Driver is already assigned to an active trip.

ACCOUNT_PENDING_APPROVAL	Fleet manager account awaits Super Manager approval.
SOS_ALREADY_ACTIVE	An SOS is already open for this trip.
RESOURCE_NOT_FOUND	Generic – entity not found.

2. Authentication (REST)

Handles login, token refresh, logout, and fleet manager signup/approval flow managed by the Super Manager.

Role: SUPER_MANAGER only

POST /auth/signup/fleet-manager

Super Manager registers a new fleet manager account

Request Body:

```
{
  "name": "Mohamed Ali",
  "email": "fleet@company.com",
  "password": "SecurePass1!",
  "phone": "01000000000",
  "businessLicense": "BL-9821",
  "fleetSize": 40,
  companyName
  companyAddress
  businessLicense
}
```

Response:

```
201 Created
{
  "success": true,
  "data": {
    "fleetManagerId": "uuid",
    "status": "PENDING_APPROVAL"
  }
}
```

Notes: Only callable by SUPER_MANAGER. The account is inactive until approved.

PATCH /auth/fleet-managers/{id}/approve

Super Manager approves a pending fleet manager account

Response:

```
200 OK
{
  "success": true,
  "data": { "fleetManagerId": "uuid", "status": "ACTIVE" }
}
```

Notes: Role: SUPER_MANAGER only. Triggers welcome email.

PATCH /auth/fleet-managers/{id}/reject

Super Manager rejects a pending fleet manager account

Request Body:

```
{ "reason": "Incomplete documentation" }
```

Response:

```
200 OK
{ "success": true, "data": { "status": "REJECTED" } }
```

Role: SUPER_MANAGER, FLEET_MANAGER, DRIVER**POST****/auth/login**

Authenticate and evaluate any user (SUPER_MANAGER, FLEET_MANAGER, DRIVER) and receive JWT tokens

Request Body:

```
{
  "email": "user@mail.com",
  "password": "123456"
}
```

Response:

```
200 OK
{
  "success": true,
  "data": {
    "accessToken": "eyJhbGciOi...",
    "refreshToken": "eyJhbGciOi...",
    "role": "FLEET_MANAGER",
    "userId": "uuid"
  }
}
```

POST**/auth/refresh**

Issue a new access token using a valid refresh token

Request Body:

```
{ "refreshToken": "eyJhbGciOi..." }
```

Response:

```
200 OK
{
  "success": true,
  "data": { "accessToken": "new_jwt_token" }
}
```

Refresh tokens are rotated on each use.

POST**/auth/logout**

Invalidate the current refresh token (server-side blacklist)

Request Body:

```
{ "refreshToken": "eyJhbGciOi..." }
```

Response:

```
200 OK
{ "success": true, "message": "Logged out successfully" }
```

Header: Authorization: Bearer <accessToken>

Role: SUPER_MANAGER, FLEET_MANAGER, DRIVER

POST /auth/forgot-password Request password-reset otp by email

Request Body:

```
{ "email": "user@mail.com" }
```

Response:

```
200 OK
{ "success": true,
  "message": "If this email is registered, a reset code has been sent.",
  "errors": [] }
```

A time-limited OTP (6-digit numeric, 10-minute TTL) is sent to the registered email address. The OTP is single-use and invalidated after use or expiry

Role: SUPER_MANAGER, FLEET_MANAGER, DRIVER

POST /auth/reset-password Verify OTP and set a new password

Request Body:

```
{ "email": "user@mail.com",
  "otp": "847291",
  "newPassword": "NewSecurePass1!" }
```

Response:

```
200 OK
{ "success": true,
  "message": "Password reset successfully. Please log in with your new password.",
  "errors": [] }
```

Error Cases

HTTP	Error Code	Meaning
400	INVALID_OTP	OTP is incorrect or has expired
400	OTP_EXPIRED	OTP TTL of 10 minutes exceeded
400	(validation)	Password does not meet complexity rules

3. Driver & Vehicle Management (REST)

Managed by FLEET_MANAGER. Drivers log in with credentials created here.

3.1 Driver Endpoints

POST

/drivers

Create a new driver account and assign to fleet

Request Body:

```
{
  "name": "Ahmed Hassan",
  "email": "ahmed@mail.com",
  "password": "InitialPass1!",
  "phone": "0100000000",
  "licenseNumber": "LIC-EG-4521",
}
```

Response:

```
201 Created
{
  "success": true,
  "data": { "driverId": "uuid", "email": "ahmed@mail.com" }
}
```

Role:Driver uses these credentials in the mobile app.

GET

/drivers/me

Get the authenticated driver's own profile (used by mobile app home screen)

Response:

```
200 OK
{
  "success": true,
  "data": {
    "driverId": "uuid",
    "name": "Ahmed Hassan",
    "phone": "0100000000",
    "plateNumber": "ABC-123",
    "currentStatus": "DRIVING",
    "fatigueAlerts": 3,
    "blindSpotAlerts": 7,
    "totalTrips": 42,
    "lastTripDuration": 480,
    "lastUpdated": "2026-02-21T09:00:00Z"
  }
}
```

Role: FLEET_MANAGER

GET /drivers/{id} Get driver profile by ID

Response:

200 OK – same schema as /drivers/me

GET /drivers List all drivers in the fleet (with optional filters)

Response:

```
200 OK
{
  "success": true,
  "data": [
    {
      "driverId": "uuid",
      "name": "...",
      "status": "IDLE",
    }
  ],
  "page": 0, "size": 20, "totalElements": 85
}
```

Query params: status, page, size

DELETE /drivers/{id} Remove a driver from the fleet

Response:

200 OK – { "success": true }

Notes: Role: FLEET_MANAGER. Cannot delete driver with an active trip.

3.2 Vehicle Endpoints

Role: FLEET_MANAGER.

POST /vehicles Register a vehicle in the fleet

Request Body:

```
{
  "plateNumber": "ABC-123",
  "model": "MAN TGX",
  "ManufactureYear": 2021,
  "License number"
}
```

Response:

```
201 Created
{ "success": true,
  "data": {
    "vehicleId": "uuid"
  }
}
```

Role: FLEET_MANAGER.

GET`/vehicles/{id}`

Get vehicle details

Response:

```
200 OK – vehicleId, plateNumber, model, year, assignmentStatus
```

Role: FLEET_MANAGER.

GET`/vehicles`

List all vehicles

Response:

```
200 OK – paginated list
```

Query params: assignmentStatus, page, size

DELETE`/vehicles/{id}`

Remove a vehicle

Response:

```
200 OK – { "success": true }
```

Keep in mind a restriction that we Cannot delete vehicles with an active trip.

هذا عيب**GET**`/vehicles/{vehicleId}/maintenance-logs`Populates the Vehicle Maintenance
CheckList widget.**Response:**

```
[ { "date": "2026-04-10", "mileage": "19,210", "details": "Engine Check", "status": "PASS" } ]
```

4. Trip Management & History (REST)

Fleet Managers create trips. Drivers control the lifecycle (start, break, end). Fleet Managers can remotely terminate.

4.1 Trip State Machine

Transition	Who triggers it
PLANNED → ONGOING	Driver: POST /trips/{id}/start
ONGOING → Paused	Driver: POST /trips/{id}/Paused
Paused → ONGOING	Driver: POST /trips/{id}/Resumed
ONGOING → FINISHED	Driver: POST /trips/{id}/end
ONGOING → CANCELLED	Fleet Manager (remote): POST /trips/{id}/cancel

POST /trips

Create a new planned trip (FLEET_MANAGER)

Request Body:

```
{
  "driverId":      "uuid",
  "vehicleId":     "uuid",
  "source":        "Cairo",
  "destination":   "Suez",
  "scheduledStartTime": "2026-02-22T06:00:00Z",
  "scheduledEndTime":  "2026-02-22T14:00:00Z"
}
```

Response:

```
201 Created
{
  "success": true,
  "data": { "tripId": "uuid", "status": "PLANNED" }
}
```

Role: DRIVER.

Notes: Validates that driver and vehicle have no concurrent active trips.

GET

/trips

Get the authenticated driver's current active trip (mobile Trip Control screen)

Response:

200 OK

```
{
  {
    "success": true,
    "data": {
      "tripId": "uuid",
      "title": "Cairo → Suez",
      "status": "ONGOING",
      "scheduledstartTime": "2026-02-22T06:05:00Z",
      "scheduledEndTime": "2026-02-22T14:00:00Z",
      "source": "Cairo",
      "destination": "Suez",
      "drivingSeconds": 3600,
      "breakSeconds": 600,
      "vehicle": { "plateNumber": "ABC-123", "model": "MAN TGX" }
    }
  },
  {
    "success": true,
    "data": {
      "tripId": "uuid",
      "title": "Cairo → Suez",
      "status": "ONGOING",
      "scheduledstartTime": "2026-02-22T06:05:00Z",
      "scheduledEndTime": "2026-02-22T14:00:00Z",
      "source": "Cairo",
      "destination": "Suez",
      "drivingSeconds": 3600,
      "breakSeconds": 600,
      "vehicle": { "plateNumber": "ABC-123", "model": "MAN TGX" }
    }
  }
}
```

Role: FLEET_MANAGER / DRIVER

GET

/trips/{tripId}

Retrieves full trip details, including related entities and expected duration calculations.

Response:

```
{
  "tripId": "TR-105",
  "title": "Aswan Trip",
  "status": "ACTIVE",
  "source": "Cairo",
  "destination": "Aswan",
  "scheduledStartTime": "2026-03-12T05:20:00Z",
  "actualStartTime": "2026-03-12T05:25:00Z",
  "estimatedArrivalTime": "2026-03-12T13:20:00Z",
  "expectedDurationHours": 8.0,
  "totalDistanceKm": 834,
  "progressPct": 64,
  "driver": {
    "driverId": "DR-1023",
    "name": "Ahmed Ali",
    "phone": "0100000000"
  },
  "vehicle": {
    "vehicleId": "Ve-305",
    "plateNumber": "ABC-123",
    "type": "Toyota Hiace"
  },
  "route": {
    "startCoords": { "lat": 30.0444, "lng": 31.2357 },
    "endCoords": { "lat": 24.0889, "lng": 32.8998 }
  }
}
```

Roles: FLEET_MANAGER, DRIVER (own trips only). // filter based on needs of the web dashboard

GET

/trips?status:active

List trips with filters

Response:

200 OK – paginated list

Role: FLEET_MANAGER

Query params: status (PLANNED|ONGOING|ON_BREAK|FINISHED|CANCELLED), driverId, vehicleId, dateFrom, dateTo, page, size

GET`/trips/{tripId}/timeline`

Get the trip event timeline for the Trip Log screen

Response:

200 OK

```
{
  "success": true,
  "message": "Trip timeline retrieved successfully",
  "errors": [],
  "data": {
    "tripLogId": "log-550e8400-e29b-41d4-a716-446655440000",
    "tripId": "trip-123e4567-e89b-12d3-a456-426614174000",
    "actualStartTime": "2026-03-03T06:05:00Z",
    "actualEndTime": "2026-03-03T13:55:00Z",
    "totalDurationHours": 7.83,
    "timeline": [
      {
        "timestamp": "2026-03-03T06:05:00Z",
        "itemType": "TRIP_STATE",
        "isAlert": false,
        "alertDetails": null,
        "details": {
          "progressState": "STARTED",
          "description": "Trip started at source warehouse",
          "location": { "lat": 30.0444, "lng": 31.2357 },
          "driverState": "NORMAL",
          "speedKmh": 0.0
        }
      },
      {
        "timestamp": "2026-03-03T06:45:00Z",
        "itemType": "TRIP_STATE",
        "isAlert": false,
        "alertDetails": null,
        "details": {
          "progressState": "ONGOING",
          "description": "Standard telemetry heartbeat",
          "location": { "lat": 30.0890, "lng": 31.3021 },
          "driverState": "NORMAL",
          "speedKmh": 65.5
        }
      },
      {
        "timestamp": "2026-03-03T07:15:30Z",
        "itemType": "EVENT",
        "isAlert": true,
```

```

    "alertDetails": {
      "alertId": "alert-999a111b-222c-333d-444e-555566667777",
      "severity": "MEDIUM",
      "acknowledged": true
    },
    "details": {
      "eventId": "evt-11112222-3333-4444-5555-666677778888",
      "eventType": "DROWSINESS",
      "description": "Elevated blink rate and yawning detected",
      "payload": {
        "blinkRatePerMinute": 18,
        "yawnCount": 4,
        "cameraConfidenceScore": 0.92
      }
    }
  },
  {
    "timestamp": "2026-03-03T08:00:00Z",
    "itemType": "TRIP_STATE",
    "isAlert": false,
    "alertDetails": null,
    "details": {
      "progressState": "ON_BREAK",
      "description": "Driver initiated scheduled break",
      "location": { "lat": 30.1500, "lng": 31.3500 },
      "driverState": "NORMAL",
      "speedKmh": 0.0
    }
  },
  {
    "timestamp": "2026-03-03T08:30:00Z",
    "itemType": "TRIP_STATE",
    "isAlert": false,
    "alertDetails": null,
    "details": {
      "progressState": "RESUMED",
      "description": "Break ended, trip resumed",
      "location": { "lat": 30.1500, "lng": 31.3500 },
      "driverState": "NORMAL",
      "speedKmh": 0.0
    }
  },
  {
    "timestamp": "2026-03-03T09:45:12Z",
    "itemType": "INCIDENT",
    "isAlert": true,
    "alertDetails": {
      "alertId": "alert-888b222c-333d-444e-555f-666677778888",
      "severity": "HIGH",
      "acknowledged": true
    }
  }

```

```

    },
    "details": {
      "incidentId": "inc-22223333-4444-5555-6666-777788889999",
      "nearMiss": true,
      "description": "Harsh braking event resulting in a near miss with a passenger vehicle.",
      "reportedAt": "2026-03-03T09:45:15Z",
      "location": "Cairo-Suez Highway, KM 45"
    }
  }
  {
    "timestamp": "2026-03-03T13:55:00Z",
    "itemType": "TRIP_STATE",
    "isAlert": false,
    "alertDetails": null,
    "details": {
      "progressState": "FINISHED",
      "description": "Arrived at destination warehouse",
      "location": { "lat": 29.9668, "lng": 32.5498 },
      "driverState": "NORMAL",
      "speedKmh": 0.0
    }
  }
]
}
}

```

These are Used for Trip Details / Trip Log screen on mobile.

POST /trips/{tripId}/start Driver starts a PLANNED trip

Request Body:

```

{
  "startLocation": { "lat": 30.0444, "lng": 31.2357 }
}

```

Response:

```

{ "success": true, "data": { "status": "ONGOING", "startedAt": "2026-03-03T06:05:00Z" },
  "message": "Trip started successfully", "errors": [] }

```

Role: DRIVER.

POST /trips/{tripId}/Paused Driver begins a rest break (ONGOING → ON_BREAK)

Request Body:

```
{ "location": { "lat": 30.100, "lng": 31.400 } }
```

Response:

```
{ "success": true, "data": { "status": "ON_BREAK", "breakStartedAt": "2026-03-03T08:00:00Z" },  
  "message": "Trip paused for break", "errors": [] }
```

Logged in timeline. Counts toward compliance break reporting.

POST**/trips/{tripId}/Resumed**

Driver ends break and resumes trip (ON_BREAK → ONGOING)

Request Body:

```
{ "location": { "lat": 30.100, "lng": 31.400 } }
```

Response:

```
{ "success": true, "data": { "status": "ONGOING", "breakDurationSeconds": 1800 }, "message": "Trip  
resumed", "errors": [] }
```

POST**/trips/{tripId}/end**

Driver ends the trip (ONGOING → FINISHED)

Request Body:

```
{  
  "endLocation": { "lat": 29.9742, "lng": 32.5498 }  
}
```

Response:

```
{ "success": true, "data": { "status": "FINISHED", "endedAt": "2026-03-03T13:55:00Z" }, "message":  
  "Trip ended successfully. Summary generation triggered.", "errors": [] }
```

Role: FLEET_MANAGER

Triggers trip summary generation.

POST**/trips/{tripId}/cancel**

Fleet Manager remotely cancels/terminates an active trip

Request Body:

```
{ "reason": "Emergency remote termination" }
```

Response:

```
{ "success": true, "data": { "status": "CANCELLED" }, "message": "Trip cancelled by management",  
  "errors": [] }
```

Role: DRIVER

POST**/trips/{tripId}/issues/maintenance****Submit Maintenance (Mobile App)****Request Body:**

```
{ "maintenanceType": "TIRE_DAMAGE",
  "payload":
    { "description": "Rear right tire blew out on the highway." }
}
```

Role: DRIVER**POST****/trips/{tripId}/issues/sos****Submit Maintenance (Mobile App)****Request Body:**

```
{ "payload":
  { "description": "Medical emergency, feeling dizzy." }
}
```

Role: DRIVER**GET****/issues?status=OPEN****Fetch All Issues (Web Dashboard)****ResponseBody:**

```
{
  "success": true,
  "data": [
    {
      "id": "req-uuid-1",
      "tripId": "TR-105",
      "category": "MAINTENANCE",
      "status": "OPEN",
      "maintenanceType": "TIRE_DAMAGE",
      "payload": { "description": "Rear right tire blew out on the highway." },
      "reportedAt": "2026-03-12T10:00:00Z"
    },
    {
      "id": "req-uuid-2",
      "tripId": "TR-108",
      "category": "SOS",
      "status": "OPEN",
      "triggerType": "MANUAL",
      "payload": { "description": "Medical emergency, feeling dizzy." },
      "reportedAt": "2026-03-12T10:15:00Z"
    }
  ]
}
```

Role: DRIVER

POST

/trips/{tripId}/incidents

POST Incident (Driver manually reports)

Request Body:

```
{
  "type": "TRAFFIC_COLLISION",
  "severity": "HIGH",
  "location": { "lat": 30.222, "lng": 31.333 },
  "payload": {
    "description": "Rear-ended a sedan at the stoplight.",
    "vehiclesInvolved": 2
  }
}
```

5. Trip filtering - Driver mobile

Role: DRIVER

timeframe convenience mapping

timeframe	Equivalent status filter
UPCOMING	status=PLANNED (scheduledStartTime in the future)
ACTIVE	status=ONGOING or status=ON_BREAK
PAST	status=FINISHED or status=CANCELLED

GET

/trips?status=timeFrame

list own trips filtered by status and/or time window

Response:

```
{
  "success": true,
  "data": [
    {
      "tripId": "uuid-1",
      "title": "Cairo → Suez",
      "status": "ONGOING",
      "scheduledStartTime": "2026-03-09T06:00:00Z",
      "scheduledEndTime": "2026-03-09T14:00:00Z",
      "actualStartTime": "2026-03-09T06:05:00Z",
      "actualEndTime": null,
      "source": "Cairo",
      "destination": "Suez",
      "vehicle": { "plateNumber": "ABC-123", "model": "MAN TGX" }
    },
    {
      "tripId": "uuid-2",
      "title": "Suez → Alexandria",
      "status": "PLANNED",
      "scheduledStartTime": "2026-03-12T08:00:00Z",
      "scheduledEndTime": "2026-03-12T18:00:00Z",
      "actualStartTime": null,
      "actualEndTime": null,
      "source": "Suez",
      "destination": "Alexandria",
      "vehicle": { "plateNumber": "XYZ-789", "model": "Volvo FH" }
    },
    {
      "tripId": "uuid-3",
      "title": "Cairo → Aswan",
      "status": "FINISHED",
      "scheduledStartTime": "2026-03-05T06:00:00Z",
```

```
    "scheduledEndTime": "2026-03-05T20:00:00Z",
    "actualStartTime": "2026-03-05T06:10:00Z",
    "actualEndTime": "2026-03-05T19:45:00Z",
    "source": "Cairo",
    "destination": "Aswan",
    "vehicle": { "plateNumber": "ABC-123", "model": "MAN TGX" }
  }
],
"page": 0,
"size": 20,
"totalElements": 42
}
```

4. Dependent (Emergency Contact) Endpoints:

Role: DRIVER

POST /drivers/{driverId}/emergency-contacts

Adds a new dependent to the driver's profile.

Request Body:

```
{
  "name": "Mahmoud Hassan",
  "phoneNumber": "01012345678",
  "relation": "Brother"
}
```

Response Body:

```
{
  "id": "dep-uuid-1",
  "name": "Mahmoud Hassan",
  "phoneNumber": "01012345678",
  "relation": "Brother"
}
```

Role: DRIVER

GET /drivers/{driverId}/emergency-contacts

Retrieves a list of the driver's emergency contacts to display on the "Emergency Contact Screen."

Response Body:

```
[
  {
    "id": "dep-uuid-1",
    "name": "Mahmoud Hassan",
    "phoneNumber": "01012345678",
    "relation": "Brother"
  },
  {
    "id": "dep-uuid-2",
    "name": "Khaled Hassan",
    "phoneNumber": "01098765432",
    "relation": "Father"
  }
]
```

5. Telemetry

Sent every 5–10 seconds from the mobile app + Raspberry Pi during an active trip. Stored as time-series data. High frequency — must be lightweight and respond fast.

It splits ingestion into **Live** (single objects pushed instantly) and **Offline Sync** (batch arrays pushed upon network recovery).

POST /trips/{tripId}/telemetry

Send periodic heartbeat snapshot

Validate this body with the appropriate needed data

Request Body:

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "timestamp": "2026-03-03T09:00:01Z",
  "location": { "lat": 30.0444, "lng": 31.2357 },
  "speedKmh": 68.4,
  "driverState": "NORMAL",
  "payload":
}
```

Response:

202 Accepted

```
{
  "success": true,
  "message": "Telemetry recorded",
  "timestamp": "2026-02-21T09:00:01Z"
}
```

driverState enum: ALERT | DROWSY | HIGH_RISK | UNCONSCIOUS. fatigueLevel: LOW | MEDIUM | HIGH.

GET /trips/{tripId}/telemetry

Retrieve telemetry history for a trip with time filters

Response:

200 OK – paginated list of telemetry snapshots

6. Events

Safety events sent immediately (not on the heartbeat schedule) when a critical condition is detected. Each event may auto-create an Alert and/or Incident.

6.1 Event Types

eventType	Trigger source
DROWSINESS	AI fatigue model – eye closure / yawning / head nod
HIGH_RISK_FATIGUE	AI model – combined high-risk threshold breached
UNCONSCIOUS	AI model – driver unresponsive classification
CRASH	IMU – acceleration magnitude exceeds crash threshold
ROLLOVER	IMU – gyroscope detects rollover angle
OBSTACLE_NEAR	Ultrasonic – blind-spot or forward proximity warning
DISTRACTION_PHONE	AI vision – phone usage detected
BREAK_EXCEEDED	System – continuous driving beyond safe limit

Role: DRIVER

POST	/trips/{tripId}/events	Sent instantly upon critical condition detection for a specific trip.
------	------------------------	---

Request Body:

```
{
  "id": "evt-uuid-1",
  "timestamp": "2026-03-03T09:05:00Z",
  "eventType": "CRASH",
  "severity": "CRITICAL",
  "payload": {
    "trigger": "MANUAL",
    "notes": "Engine failure"
  }
}
```

Response:

```
201 Created
{
  "success": true,
  "data": {
    "eventId": "uuid",
    "alertCreated": true,
  }
}
```

GET`/events`

List events with filters

Response:

200 OK – paginated list

Query params: tripId, eventType, severity, from, to, page, size

GET`/events/{id}`

Get event details

Response:

200 OK – full event object

Role: DRIVER**POST**`/sync/offline-data`

Triggered only when the mobile app reconnects after a network drop..

Request Body:

```
{ "telemetryRecords": [...], "events": [...] }
```

Response:

202 Accepted

7. Alerts

Alerts are auto-created from Events. They appear in the driver's mobile app and the fleet manager dashboard.

GET

`/alerts/?status=UNACKNOWLEDGED`

List alerts with filters

Response:

```
[ { "alertId": "uuid", "eventType": "FATIGUE", "severity": "HIGH", "driverName": "Ahmed Ali" } ]
```

Query params: tripId, driverId, severity, acknowledged, eventType, page, size

ROLE: FLEET MANAGER

GET

`/alerts/{id}`

Get alert details

Response:

```
{
  "success": true,
  "data": {
    "alertId": "uuid",
    "severity": "HIGH",
    "eventType": "DROWSINESS",
    "acknowledged": false,
    "readByManager": false,
    "createdAt": "2026-03-03T07:15:30Z",
    "ackedAt": null,

    "trip": {
      "tripId": "uuid",
      "title": "Cairo → Suez",
      "status": "ONGOING",
      "scheduledStartTime": "2026-03-03T06:00:00Z",
      "scheduledEndTime": "2026-03-03T14:00:00Z",
      "source": "Cairo",
      "destination": "Suez"
    },

    "driver": {
      "driverId": "uuid",
      "name": "Ahmed Hassan",
      "phone": "0100000000",
      "currentState": "DROWSY"
    },

    "vehicle": {
      "vehicleId": "uuid",
      "plateNumber": "ABC-123",
      "model": "MAN TGX"
    }
  },
}
```

```
{
  "event": {
    "eventId": "uuid",
    "eventType": "DROWSINESS",
    "timestamp": "2026-03-03T07:15:30Z",
    "payload": {
      "blinkRatePerMinute": 18,
      "yawnCount": 4,
      "cameraConfidenceScore": 0.92
    }
  }
}
```

POST /alerts/{alertId}/ack

Driver or fleet manager acknowledges an alert (FR-14)

Role: FLEET_MANAGER

Response:

```
{ "success": true, "data": { "acknowledged": true, "ackedAt": "2026-03-03T09:20:00Z" },
  "message": "Alert acknowledged", "errors": [] }
```

POST /alerts/{alertId}/read

Fleet manager marks alert notification as seen/read

Role: FLEET_MANAGER

No request needed, as it will be Called automatically when the manager opens or views the alert detail screen

Response:

```
{ "success": true,
  "data": {
    "alertId": "uuid",
    "readByManager": true,
    "readAt": "2026-03-03T07:20:00Z"
  },
  "message": "Alert marked as read",
  "errors": []
}
```

GET`/issues?resolved=false&type=SOS`

Lists SOS and Maintenance tickets that require human intervention.

Response:

```
[ { "issueId": "uuid", "tripId": "uuid", "type": "SOS", "resolved": "false", "createdAt": "..." } ]
```

POST`/issues/{issueId}/resolve`

Marks an SOS or maintenance ticket as completed.

Request Body:

```
{ "resolutionNote": "Dispatched maintenance team to location." }
```

GET`GET /incidents`

Lists historical accidents/near-misses for the Alerts & Incidents page

Response:

```
[ { "incidentId": "uuid", "type": "COLLISION", "severity": "HIGH", "description": "..." } ]
```

12. Dashboard & Fleet Monitoring (WebSockets)

This page gives a high-level overview of the live operations, driver states, and active alerts.

Live Map Data (WebSockets)

SUBSCRIBE /topic/fleet/{fleetId}/telemetry

Stream live GPS coordinates, speed, and status to smoothly move the truck icons on the map.

Live Alerts (WebSockets)

SUBSCRIBE /topic/fleet/{fleetId}/alerts

Pushes an **Alert** JSON object the exact millisecond the backend creates an incident or issue request..

GET /dashboard/{fleetId}/alerts-summary

Aggregated alert statistics for the fleet manager

Response:

```
200 OK
{
  "success": true,
  "data": {
    "totalToday": 24,
    "critical": 3,
    "unacknowledged": 11,
    "byType": {
      "DROWSINESS": 12,
      "OBSTACLE_NEAR": 8,
      "CRASH": 1
    }
  }
}
```

Get Fleet Status KPIs

GET /dashboard/{fleetId}/fleet-status

Online/offline/idle/driving status of all drivers

Response:

```
{
  "activeVehicles": 10,
  "activeVehiclesChangePct": 5,
  "activeTrips": 4,
  "activeTripsChangePct": 8
}
```

Get Active Alerts Widget

GET

/alerts?status=UNACKNOWLEDGED&limit=5

Get active alert filtered

Response:

Array of recent alerts (e.g., Fatigue, Blind Spot, SOS) to show in the bottom right corner.

Get Drivers Status KPIs

GET

/dashboard/drivers-status

Get active Driver status

Response:

```
{
  "alert": 22,
  "drowsy": 7,
  "highRisk": 3
}
```

GET

/dashboard/active-trips

Populates the "Trips Summary Table" and initializes the Live Map

Response:

```
[ { "tripId": "TR-105", "title": "Aswan Trip", "driverName": "Ahmed Ali", "vehiclePlate":
"Ve-305", "status": "ACTIVE", "progressPct": 64, "location": { "lat": 24.0889, "lng": 32.8998 } }
```

GET /locations/nearest-help

- **Description:** Populates the "Nearest Help" widget on the map view.
- **Query Params:** ?lat=24.0889&lng=32.8998&radius=20
- **Response Data:** [{ "name": "Aswan Hospital", "type": "HOSPITAL", "distanceKm": 20 }]

13. Report Generation

Auto-generated PDF reports for compliance, insurance, and post-incident review.

GET `/reports/trip/{tripId}`

Generate and download a full trip safety PDF report

Response:

200 OK – Content-Type: application/pdf (binary download)

Notes: Includes: route, timeline, events, driver state log, break periods, alerts, incidents.

GET `/reports/driver/{driverId}`

Generate driver safety performance report over a date range

Response:

200 OK – Content-Type: application/pdf

GET `/reports/kpis?dateRange=LAST_YEAR`

Populates the top row of the Reports & Analytics page.

Response:

```
{ "tripsCompleted": 294, "fatigueAlerts": 27, "sosAlerts": 8 }
```

GET `/reports/trips/{reportId}/download`

Downloads the actual file when the manager clicks the "Download" link

Response:

Binary `application/pdf`

14. Notification Settings

GET

`/drivers/{driverId}/notifications`

Downloads the actual file when the manager clicks the "Download" link

JSON

```
{
  "success": true,
  "data": [
    {
      "id": "notif-uuid-101",
      "type": "TRIP_ASSIGNED",
      "title": "New Trip Assigned",
      "message": "You have been assigned to the Cairo-Aswan route.",
      "isRead": false,
      "createdAt": "2026-03-12T10:00:00Z",
      "details": {
        "tripId": "TR-122",
        "vehiclePlate": "ABC-123",
        "scheduledStartTime": "2026-03-13T06:00:00Z"
      }
    },
    {
      "id": "notif-uuid-102",
      "type": "SAFETY_ALERT",
      "title": "Rest Break Required",
      "message": "You are approaching your maximum continuous driving limit.",
      "isRead": true,
      "createdAt": "2026-03-11T14:30:00Z",
      "details": {
        "alertId": "alert-uuid-444",
        "severity": "HIGH",
        "recommendedAction": "PULL_OVER"
      }
    },
    {
      "id": "notif-uuid-103",
      "type": "SYSTEM_BROADCAST",
      "title": "App Update Available",
      "message": "Please update your Udjat app to version 2.1 for the latest GPS improvements.",
      "isRead": true,
      "createdAt": "2026-03-10T09:00:00Z",
      "details": {
```

```
    "actionUrl": "https://play.google.com/store/apps/details?id=com.udjat.driver"  
  }  
}  
],  
"message": "Notifications retrieved successfully",  
"errors": []  
}
```

7. Web Dashboard & Real-Time Monitoring (WebSockets)

Replaces continuous REST polling. The Web Dashboard connects to these channels to receive instant updates pushed by the backend [EventBus](#).

Connection Protocol

- **Type:** STOMP over WebSockets.
- **Handshake:** Authenticated via JWT token in the connection header.

Channels (Topics)

- **SUBSCRIBE /topic/fleet/{fleetManagerId}/telemetry**
 - **Payload:** Streams the updated [TripState](#) JSON object every time a truck syncs a new telemetry heartbeat. Used to smoothly move truck icons on the live map.
- **SUBSCRIBE /topic/fleet/{fleetManagerId}/alerts**
 - **Payload:** Pushes an [Alert](#) JSON object the exact millisecond the backend [AlertProcessor](#) generates it (e.g., when a CRASH or UNCONSCIOUS event is processed). Triggers UI pop-ups and red map indicators.

Appendix: Enum Reference

Enum	Values
Role	SUPER_MANAGER FLEET_MANAGER DRIVER
Trip Status	PLANNED ONGOING ON_BREAK FINISHED CANCELLED
Driver State	ALERT DROWSY HIGH_RISK UNCONSCIOUS
Fatigue Level	LOW MEDIUM HIGH
Event Type	DROWSINESS HIGH_RISK_FATIGUE UNCONSCIOUS CRASH ROLLOVER OBSTACLE_NEAR DISTRACTION_PHONE BREAK_EXCEEDED
Severity	LOW MEDIUM HIGH CRITICAL
SOS Trigger	MANUAL AUTOMATIC CRASH_DETECTED
SOS Status	ACTIVE RESOLVED
Driver Status	ONLINE OFFLINE DRIVING ON_BREAK IDLE
Issue Category	ENGINE BRAKE TIRE ELECTRICAL VIBRATION OTHER
License Type	HEAVY_TRUCK LIGHT_TRUCK GENERAL
Fleet Manager Status	PENDING_APPROVAL ACTIVE REJECTED SUSPENDED